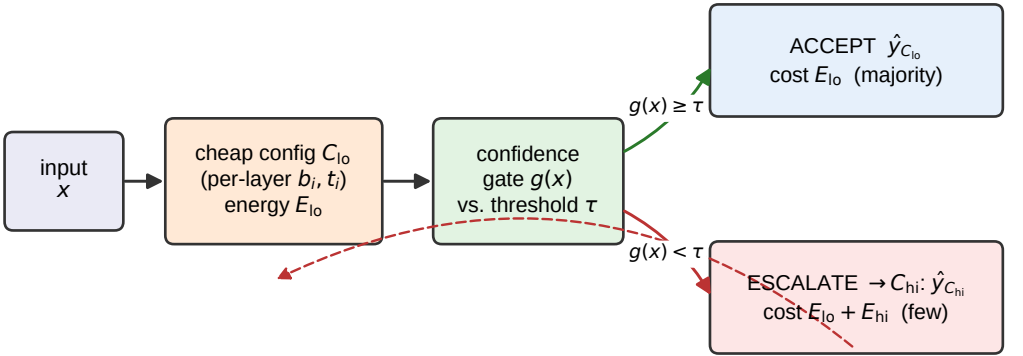
Expected energy per sample: $\mathbb{E}[E] = E_{\text{lo}} + p_{\text{esc}}\, E_{\text{hi}}, \quad p_{\text{esc}} = \Pr[g(x) < \tau]$



escalated input re-processed at higher precision (cheap pass already paid)